A Coulomb–Newton Cosmology

Based on a Proton–Electron Model (RealQM / RealNucleus)

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Abstract

We sketch **RealUniv**, an exploratory cosmology built from two forces that share *one field equation.* Both Coulomb's electrostatics and Newton's gravitation are Poisson equations, $\nabla^2\varphi = \pm(\text{source})/\text{constant}$, differing only in a sign (like charges repel, like masses attract) and a strength. This **two-valued Poisson** structure has **two sources** — charge and mass/energy — and **two constants**, one per force. **Coulomb, on charge, rules the small**: charge of two spatial extents — small dense positive (protons) and large diffuse negative (electrons) — builds nuclei (RealNucleus) and atoms (RealQM) by pure Coulomb geometry, with no strong force and no elementary neutron. **Newton, on the energy of that matter, rules the large**: since mass is energy content (E/c^2, a *measured* identity), the energy density $\rho \geq 0$ of Coulombic matter sources gravitation through a *two-signed* Poisson equation, $\pm\frac{1}{4\pi G}\nabla^2\varphi_g = \rho$, giving the gravitational potential regions of *positive and negative curvature* (positive and negative mass). Under the mirror force-rule — like masses attract, unlike repel, positive and negative behaving identically — the large scale splits into *interfoliated chunks of like mass*: the cosmic web of positive-mass filaments and clusters, and negative-mass **voids** (empty of ordinary matter) whose repulsion of it acts as dark energy. The two forces run **in parallel** — same equation, opposite sign, own constant — and we make *no* claim that one is derived from the other, nor that mass equals size; whether the two constants ultimately reduce to one is left as an open question (Section 11). The construction is strictly non-relativistic — Coulomb below, Newton above — so c enters only the electromagnetic sector: a local speed limit for charges and light, but none for the large-scale gravitational motion of neutral bodies (two galaxies may recede at relative speed $> c$, as observed). We state the picture, note that it *reproduces* the primordial helium fraction as a consistency (with the observed n–p gap E_* and the standard freeze-out supplied, not derived), and set out its load-bearing open problems — chiefly *what sets the two constants, and whether they are at bottom one.*

Status. An exploratory proposal, not established cosmology. It departs from the Standard Model and ΛCDM and must eventually be reconciled with them; the open problems of Section 11 are load-bearing. Read it as "what follows *if*."

Keywords: cosmology; two-valued Poisson; Coulomb and Newton; proton–electron model of the nucleus; dark energy; primordial nucleosynthesis; RealQM.

1 Introduction

Cosmology is unavoidably speculative at its foundations. The standard account — hot Big Bang, cosmic inflation, ΛCDM — fits the data only by invoking ingredients that are inferred rather than seen: an initial singularity, an inflationary epoch, dark matter, and dark energy, none of them observed in a laboratory, each introduced to close a gap. In a subject resting on such foundations it is legitimate, and useful, to ask what a *simple, rational, non-standard* alternative would look like — one built not from new fields and new epochs but from the most ordinary ingredients physics already has: **electron–proton Coulomb interaction and Newtonian gravitation**. This note develops one such model.

The RealQM program formulates quantum mechanics as multiphase 3D continuum mechanics — unit-charge densities on non-overlapping domains separated by free boundaries, minimising the Coulomb energy [1]. Its nuclear extension, RealNucleus, revives the pre-1932 proton–electron picture (neutron = proton + electron) and reproduces nuclear binding from pure Coulomb interactions, with no strong-force postulate [2]. This note asks what cosmology results if the *same* ingredients are taken as the starting point of the universe, together with ordinary Newtonian gravitation.

The answer, which we call **RealUniv**, is a **two-tier** cosmology carried by **two mirror forces**. Coulomb and Newton are the same field equation — Poisson's — with opposite sign and a different strength (Section 2). **Coulomb, acting on charge**, shapes the small scale (charge → nuclei → atoms, all of RealQM); **Newton, acting on the energy of that charged matter**, shapes the large scale (the cosmic web, with a repulsive negative-mass sector as dark energy). The two run in parallel, one force per scale; we do not claim either is derived from the other. We keep strictly to what Coulomb and Newton can say, mark the extreme regimes (strong gravity, the highest energies) where more would be needed, and keep the open problems — above all whether the two forces are ultimately one — explicit.

2 The machinery: two-valued Poisson, opposite force-signs

Both forces share *one* structure and differ in a single sign. Each is the gradient of a potential obeying Poisson's equation with a two-valued (±) source,

$$\text{electric:} \quad \rho_c = -\varepsilon_0\,\nabla^2\varphi_E, \qquad\qquad \text{gravitational:} \quad \rho_m = \frac{1}{4\pi G}\,\nabla^2\varphi_g, \tag{1}$$

the force being $-(\text{charge})\,\nabla\varphi$ in each case. In this cosmology the Coulomb source is a fluctuation of the electric potential (Section 3) and the Newton source is the energy of the charged matter (Section 6); the two forces run in parallel and the *machinery* is common. The one structural difference is the relative sign, and with it the force law:

	like sources	unlike sources
electric (charge)	**repel**	attract
gravitational (mass)	**attract**	repel

Concretely there are two independent binary signs — the source $\pm\nabla^2\varphi$ (two-valued) and the force $\mathbf{F} = \pm\nabla\varphi$ (repulsive or attractive for like sources) — and their *four* combinations are exactly the four "charges" the two forces couple to: the repulsive-for-like sign is electromagnetism, its ± source the ±charge of the small scale; the attractive-for-like sign is gravitation, its ± source the ± mass/energy of the large scale. One operator and two signs thus yield {±charge, ±mass}.

This single sign difference generates the two tiers. Because like charges *repel* and unlike attract, matter neutralises into ± pairs, charge cancels, and electromagnetism *screens* — it stays local and small-scale. Because like masses *attract*, mass *clumps* and accumulates, growing without bound into large-scale structure; and because unlike masses repel, positive- and negative-mass regions *separate* into the interfoliated web and voids of Section 6. The scale separation is thus a *consequence* of the force-sign, not a separate assumption: small-scale electromagnetism and large-scale gravitation are one two-valued Poisson structure, read with two signs. (This is the static Poisson level; the electromagnetic sector carries c and radiation, and ±-mass gravitational dynamics has its own subtleties — both belonging to the extreme regimes outside the Coulomb–Newton scope kept here.)

A second asymmetry — in the *direction* the Poisson relation is used — fixes the *scale* of each force. For electromagnetism the *potential* is primary: the seed is a small, *small-scale* (short-wavelength) oscillation of φ_E — not a smooth ripple — and charge is obtained by *differentiating*, $\rho_c = -\varepsilon_0 \nabla^2 \varphi_E$. The Laplacian is a *local* operator that *magnifies* short wavelengths ($\rho_k \propto k^2 \varphi_k$), so precisely because the seed is small-scale (large k) it is inflated, despite its tiny amplitude, into sharp, small-scale $\pm$ charges. For gravitation the *source* is primary — the mass/energy of the charged world — and the potential is obtained by *integrating*, $\varphi_g = 4\pi G\, \nabla^{-2} \rho_m$. The inverse Laplacian is a *global* operator (convolution with the $1/r$ Green's function) that *smooths*, suppressing short wavelengths ($\varphi_k \propto \rho_k / k^2$) and building a broad, large-scale potential. So the same Poisson relation, run forward makes small-scale electromagnetism and run backward makes large-scale gravitation: **the Laplacian acts locally and magnifies; the inverse Laplacian acts globally and smooths** — and that is why charge is small and gravity large.

3 The arena and the sources

The arena is a plain **3D Euclidean coordinate space** — absolute space, simply given, with no physical medium. The only operator needed is the Laplacian ∇^2. The small-scale structure begins with a fluctuation of the **electric potential** φ_E, the source of the Coulomb sector. Charge is not postulated separately; it is the curvature of that potential,

$$\rho_{\text{charge}} = -\varepsilon_0\, \nabla^2 \varphi_E, \tag{2}$$

with two immediate consequences. **Both signs:** ρ_{charge} is positive where φ_E is concave, negative where convex. **Net zero:** $\int \nabla^2 \varphi_E\, dV$ is a surface term that vanishes, so the universe carries **zero net charge** automatically — equal numbers of protons and electrons, with no baryogenesis (and, once the neutron is read as a proton–electron pair, in the far stronger *local* form of Section 4). And because the Laplacian multiplies each Fourier mode by k^2 ($\rho_k \propto k^2 \varphi_k$), a small, *small-scale* (short-wavelength) oscillation of φ_E — not a smooth ripple — becomes a large density contrast: the seed is tiny in amplitude but short in wavelength, and it is the short wavelength (large k) that the Laplacian amplifies into strong charge.

This fluctuation is the source of the *first* force, Coulomb. The *second* force, Newton, has its own source — the mass/energy of the matter that Coulomb then builds (Section 6) — and its own Poisson equation and constant. The two run **in parallel**, one force per scale (Section 2). We do *not* assert that the gravitational source is nothing but the energy of the charge structures, i.e. that the two sectors collapse to a single seeded potential: that reduction — which would fold the two constants into one — is an appealing possibility but an *open* one, taken up in Section 11, not a premise here.

4 The small scale: RealQM and RealNucleus

The seed, inflated by the Laplacian, produces charge of both signs and, decisively, of *different spatial extent*: small dense positive charge and large diffuse negative charge. These are the **proton** and the **electron**, and the whole charged world is built from them by Coulomb geometry.

Energy, not mass — and its relation to size. In RealQM a particle is a charge cloud whose size is fixed by the balance of a localisation (kinetic) cost against Coulomb attraction, and its inertia is simply its energy content, E/c^2 — a *measured* identity (the Penning trap, the nuclear mass defect), not a relativistic postulate [3]. We therefore invoke no primitive "mass" at all: charged matter carries *energy*, and it is energy that gravitates (Section 6). A *contracted* cloud costs more localisation and Coulomb energy, so *qualitatively* small clouds are energy-dense — the small proton is the heavy one, the large electron the light one. Whether size fixes the energy *quantitatively* is another matter: the exact size–energy map is not the naive inverse one — the proton is more compact than that would make it, by about a factor of ten (Section 11). So energy and size are closely *related*, not *identified*; we assert only the measured part (inertia is energy) and leave the map open. Nothing below depends on closing it — the cosmology needs only that gravity couples to energy.

No overlap, no self-interaction: self-repulsion is kinetic energy. RealQM charge clouds do not overlap — each electron occupies its own domain with a free boundary — and carry *no self-interaction*: there is no Coulomb self-repulsion term. The role such self-repulsion would play, resisting the collapse of a charge to a point, is carried instead by the *localisation energy* $\frac{1}{2}\int|\nabla\psi|^2$: compressing a cloud sharpens its gradients and costs kinetic energy, so it resists compression and settles at a finite size (localisation cost against Coulomb attraction). A free electron, unconfined, therefore spreads; a bound one takes the size where localisation balances attraction. In RealQM, **self-repulsion appears as kinetic energy**.

Charge is quantized: ± 1 the only possibility. This same structure forces charge to come in identical units — with no monopole and no compact gauge group imported. To exclude an electron's Coulomb energy with *itself*, the "self" must be well defined: there must be an *indivisible* entity whose interior is left out of the Coulomb sum. A divisible continuum of charge would make "self" ambiguous — is half a cloud a self? do the two halves repel? — and the no-self-interaction prescription ill-posed. So removing self-repulsion *requires* indivisible units. An indivisible unit is identified by its *spatial occupation* — it is precisely what fills one domain — and for all electrons, and equally all protons, to be the *same* entity (identity), each such domain must carry the same charge. Discreteness and universality then follow at once: every electron carries one unit -1 and every proton $+1$, the two signs of the seed (Section 3), and no intermediate charge can exist. The single premise is that the no-self-interaction law is one *universal* law (the same "self" for every electron), not an ad-hoc per-blob subtraction; granted that, quantization is a *consequence*, not a postulate. This is where the Standard Model must *import* the result — anomaly cancellation fixes only the $\pm$ *ratio* of charges, and true quantization needs a Dirac monopole or a compact unifying group, both unobserved. What this argument does *not* fix is the *magnitude* of the unit (why the quantum has the size it does — a scale question, sibling to the nuclear scale of Section 11), nor the fractional charges of confined quarks, which lie outside RealQM's no-substructure scope.

Nuclei and atoms, one mechanism at two scales. With a neutron taken as $p+e$, a neutral atom (A, Z) holds A protons and A electrons: $A-Z$ *compact* inside the nucleus and Z *expanded* in orbitals. The electron occupies two size-phases, its extent adapting to the Coulomb well: compact in a deep well (inside a nucleus, $\sim$fm, $\sim$MeV) and expanded in a shallow one (atomic, $\sim$Å, $\sim$eV). Chemistry and nuclear physics are then one Coulomb free-boundary mechanism at two scales: a molecule is nuclei glued by expanded (valence) electrons; a nucleus is protons glued by compact electrons. RealNucleus reproduces nuclear binding this way — the He-4/deuteron binding ratio comes out 13.1 against an experimental 12.7, from Coulomb geometry alone [2], and light-nucleus fusion (hydrogen burning) appears as a rearrangement of protons and electrons, not the action of a separate force.

Neutrality is local, not merely cosmic. Because a neutron *is* a bound proton–electron pair, every nucleus (A, Z) carries exactly A protons and A electrons, and $N_e = N_p$ is enforced *atom by atom, by construction* — not as a distant global sum. This is a far stronger statement than cosmic neutrality. Global neutrality asks only that the total charge of the universe vanish; it would tolerate vast regions of net positive and net negative charge cancelling only in the grand total. Local neutrality forbids exactly that: it holds at every atom, and local neutrality *implies* the global kind while the converse fails — so the local form is logically much the stronger.

It is also the form actually put to the test: laboratory bulk-matter experiments bound the residual per-atom charge to $|q_p + q_e| \lesssim 10^{-21}\, e$, a direct and stringent constraint, whereas cosmic net-charge limits are indirect inferences. In the Standard Model this exact local balance needs two separate ingredients — charge quantization (from anomaly cancellation) and an *assumed* neutral initial condition — with the neutron, a distinct neutral particle, contributing nothing to the charge ledger and its abundance left to weak freeze-out. Here the same balance is automatic twice over: globally from the net-zero seed ($\int \nabla^2 \varphi_E = 0$, Section 3), and locally because the neutron cannot unbalance a count of which it *is* the matched pair. What remains owed is charge *quantization* — why each cloud carries exactly one e — which rides on the discreteness of the RealQM domains, a separate claim.

The nuclear/atomic scale. What sets the two sizes can be read from charge and the proton's finite size R_p alone, with *no mass and no c*. An electron attracted to one proton in vacuum must taper to zero at infinity; the localisation cost then forces a large cloud of size the atomic $a_0 = \hbar^2/me^2 \approx 0.5\,\text{Å}$. An electron caged among protons at a multiphase free boundary need *not* taper to zero — its density hands off to the proton density — so a flat caged density has zero localisation cost, and it sits at the cage (proton) size $R_p \approx$ a few fm, bound by Coulomb ($\sim$MeV at fm). The size ratio is therefore

$$\frac{a_0}{R_p} \;\approx\; \frac{1}{\alpha^2} \;\approx\; 2 \times 10^4, \tag{3}$$

so the fine-structure constant appears here as a pure *geometric* ratio of the two scales, $\alpha^2 = R_p/a_0$, the ratio of nuclear to atomic size — not a coupling constant. Empirically the H atom ($\approx a_0 = 0.53\,\text{Å}$) and the deuteron nucleus ($\approx 2.1\,\text{fm}$) give a ratio $\approx 1/25{,}000$, matching $\alpha^2 \approx 1/18{,}800$ to order of magnitude. *Why* the proton is $\sim$fm — what fixes R_p — remains the central open problem.

A ceiling on the scale, without relativity. That one open number is not unbounded: RealQM sets a floor under R_p. RealQM describes a *fixed number of persistent charge clouds*, a description valid only while no energy in play reaches an electron's own energy content E_e (its rest energy, $\sim$0.5 MeV — a directly measured quantity, Section 5, not a relativistic postulate). A compact electron bound by Coulomb at the nuclear scale has binding $\sim e^2/R_p$; requiring it to stay below E_e,

$$\frac{e^2}{R_p} \lesssim E_e \qquad \Longleftrightarrow \qquad R_p \gtrsim \frac{e^2}{E_e}, \tag{4}$$

puts a floor under the proton size. Note the criterion is a plain comparison of two measured energies — the binding must be less than an electron's worth — and needs *no* appeal to α (the $a_0/R_p \lesssim 1/\alpha^2$ form above is only this same statement re-expressed as a ratio to the atomic size). The reason crossing the floor would break the picture — pair creation — is relativistic and lies outside RealQM, but the *boundary itself* is stated entirely within it: below it, clouds no longer persist. Physically it says RealQM is safe as long as it deals with far less than an electron's worth of energy, and strains as that budget is approached. And nature sits essentially *at* the floor: at the deuteron radius ($\sim$2.1 fm) the Coulomb binding $e^2/R_p \approx 0.7$ MeV already exceeds $E_e \approx 0.5$ MeV — so the nuclear scale saturates the theory's own validity limit, which is why the nuclear regime, and β-decay in particular, is where the static-cloud description is most stretched.

5 Inertia is energy; the local speed limit

Throughout, inertia is the energy content: $\mathbf{p} = (E/c^2)\mathbf{v}$, the measured relation of the Penning trap [3], and Newton's second law $\mathbf{F} = \dot{\mathbf{p}}$ governs the response. In the small, the force is electromagnetic and couples to *charge*; in the large, it is gravity and couples to *mass* $= E/c^2$. One mechanics, one inertia, two forces reading two properties of the same matter.

The constant c enters *only* the electromagnetic sector — through the field momentum of a moving charge, Thomson's electromagnetic mass. Consequently there is a **local speed limit**: a charge, or a pulse of light, cannot overtake light *through* the local matter and fields, because feeding it energy only increases its inertia E/c^2 without bound as $v \to c$. But **Newtonian gravitation is c-free** — instantaneous, with no c in $\rho = \nabla^2\varphi_g$ — so it imposes *no* speed limit on the large-scale motion of neutral bodies. Two galaxies may approach, or recede, at relative speed *greater than* c; this is a global relative velocity of gravitating matter, not a local signal, and it violates no local causality (no local frame sees anything pass light through it). The speed limit is thus a property of the RealQM/Maxwell sector, not of gravity — just as in observational cosmology distant galaxies recede faster than light without any local law being broken. (How Newtonian mechanics extends to relatively moving frames — where c and its kinematics would enter — is a separate matter, taken up in *Many-Minds Relativity* [4], and lies outside the non-extreme scope kept here.)

6 The large scale: Newtonian gravitation

The mass/energy density of matter — protons, electrons, their binding and radiation — is ρ, **positive throughout** ($\rho \geq 0$). This is the source of the *second* force, Newton, through its Poisson equation, and the gravitational potential φ_g comes in **two forms**, its Laplacian taking either sign against the one positive source:

$$\pm\frac{1}{4\pi G}\nabla^2\varphi_g \;=\; \rho \;\geq\; 0. \tag{5}$$

The $+$ form is a *well* ($\nabla^2\varphi_g > 0$, positive mass), the $-$ form an *anti-well* ($\nabla^2\varphi_g < 0$, negative mass); the $\pm$ mass is *formal* — the sign carried by the curvature of φ_g, not a negative energy, the source ρ being positive. The gravitational force-rule is the mirror of Coulomb's — **like masses attract, unlike masses repel** — and positive and negative mass behave *identically* under it: each attracts its own kind and repels the other; neither is distinguished. So the large scale **splits symmetrically into chunks of like mass** — positive-mass chunks and negative-mass chunks — which clump internally and repel one another apart. We, being made of *positive* matter, see the positive chunks as the luminous **cosmic web** of filaments and clusters; the negative-mass chunks, empty of ordinary matter and repelling it, appear to us as **voids**. The apparent voids are therefore *not empty but regions of negative mass*, and their repulsion of the positive matter is what shows up as **dark energy**.

Both sectors are the two signs of the one gravitational Poisson equation, sourced by the single positive ρ, and behave symmetrically: each clumps internally (like attracts like) while the two repel and **separate** (unlike repels) — so the universe interfoliates into positive- and negative-mass chunks, the negative chunks being what the positive sector perceives as voids. This is the Newton sector's own structure — the large-scale mirror of the Coulomb sector below (same equation, opposite sign, own constant G). Whether ρ is *nothing but* the energy of the charge structures — so that the two sectors would share a single origin — is the reductive question left open in Section 11; here we take gravity as the second force in its own right, acting on the energy of the matter Coulomb builds.

A 2D test: cosmic-web morphology. A mass-conserving 2D simulation of a $\pm$mass field $\rho = \nabla^2\varphi$ from a random potential, evolved as compressible self-gravitating flow with same-sign attraction and opposite-sign repulsion, develops the morphology of observed large-scale structure (Figure 1): intermixed positive- and negative-mass filamentary webs, mass concentrated at nodes, separated by large voids — the voids here *actively driven* by negative-mass repulsion rather than merely evacuated by gravity. A qualitative match (a 2D toy); quantitative tests (void-size distribution, correlation function) remain to be done.

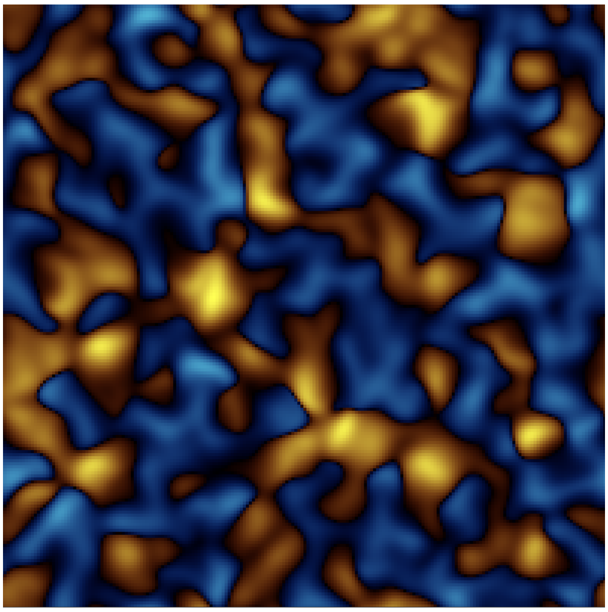

Figure 1: 2D self-gravitating ±mass from $\rho = \nabla^2\varphi$: positive- (warm) and negative- (cool) mass webs with concentrated nodes and large voids — cosmic-web morphology, voids driven by negative-mass repulsion. Mass-conserving; periodic box.

7 Dark energy and inflation from the same machinery

Two central features of modern cosmology appear here as consequences of the *same* two-valued Poisson machinery — one from each sector (a Coulomb-side amplification of the charge field, a Newton-side repulsive component) — an economy worth stating, and honestly bounding.

Dark energy. The negative-mass sector of Section 6 is what we perceive as the **voids**: regions of negative mass, empty of ordinary (positive) matter and repelling it (unlike masses repel). That repulsion pushes the positive matter apart and drives the voids to grow — a built-in **dark-energy mechanism**, not put in by hand, but simply the negative-mass sign of the two-signed gravitational Poisson equation. So the apparent voids are not empty space accelerating for want of a cause; they are the second gravitational sign doing its symmetric work. What is *not* yet derived is its magnitude, or that the effective equation of state is $w \approx -1$; the mechanism is qualitative.

That a *negative-mass* sector might unify the dark components has been proposed before, most prominently by Farnes [7], whose single negative-mass fluid — sustained by continuous matter creation — aims to account for both dark matter and dark energy within a modified ΛCDM framework (a proposal that has itself drawn debate). RealUniv shares the core intuition that negative mass drives cosmic acceleration, but differs at the root. There the negative mass is a *primitive* fluid, with genuinely negative inertial mass and runaway motion, requiring a matter-creation term and retaining the expanding-space ΛCDM scaffold; here it is the *emergent, formal* anti-well sign of a gravitational potential sourced entirely by ordinary *positive* energy — no negative inertial mass, no creation term, no Big Bang, and no separate dark-matter component at all (cosmic structure is the ordinary Coulombic mass of Section 6). The two proposals converge on negative mass as the key to the dark sector and diverge on whether it is fundamental or emergent.

They differ, too, in scope. Farnes advances his negative masses avowedly as a *toy model* — a phenomenological component grafted onto ΛCDM to demonstrate that negative mass *can* mimic the dark sector, illustrated by toy N-body simulations. RealUniv instead offers the negative sector not as a separate fluid but as the *anti-well sign* of the gravitational Poisson equation itself — part of the same two-valued machinery that gives the cosmic web, sourced by ordinary positive energy, not bolted onto a standard scaffold. The ambition is thus a first-principles cosmology rather than a dark-sector patch. Honesty requires the counterweight: on the dark sector *specifically*, RealUniv is so far only qualitative (the 2D toy of Section 6) and *less* quantitatively developed there than Farnes's simulations. RealUniv's claim to be the fuller model rests on integrative scope — one common machinery for matter (Coulomb) and the dark sector (Newton) — not on validated dark-sector dynamics, which remain to be built.

Inflation. The Laplacian amplifies short wavelengths, $\rho_k \propto k^2\varphi_k$, so a tiny small-scale oscillation of the potential is magnified into a large density contrast — an **"inflation" of the density field**. This is amplification of the *field*, not the exponential expansion of *space* of the standard inflationary paradigm (which addresses the horizon and flatness problems); the two must not be conflated.

Because $\rho = \nabla^2\varphi$ is k^2-weighted, the amplification is short-wavelength dominated: the seed comes out as *small-scale* charge structure. In this model that is not a spectral defect to be observed in the sky but the very mechanism by which the initial fluctuation becomes a **proton–electron universe** — nuclei, atoms, bound matter at small scales (Section 4). One might object that a k^2-weighted ("blue") density spectrum conflicts with the near scale-invariant, slightly red primordial spectrum ($n_s \approx 0.96$) inferred from the CMB. But that inference is internal to the hot Big Bang: it reads the CMB as the red-shifted *afterglow* of a hot dense phase and reconstructs a primordial density spectrum through recombination and a transfer function. This model has no Big Bang and no afterglow phase; it makes no claim about the CMB and requires none. The n_s comparison therefore presupposes a framework the model does not adopt, and the short-wavelength weighting is read here as a feature — the origin of small-scale matter — rather than a conflict. The interpretation of the CMB itself we regard as a separate, open observational question outside the scope of a small-scale-structure cosmology.

So from the one common machinery the model contains natural *candidates* for both dark energy (the negative-mass sign of the Newton equation) and a fluctuation amplification ("inflation of the density field," the k^2-weighting of the Coulomb source) — an economy that a separate-mechanism cosmology does not have — while neither is yet quantitatively validated.

8 The cosmic sequence and the helium fraction

The two sectors exchange energy in *both* directions, and the cosmic sequence is that exchange in time. *Upward*, the energy of the charged world is mass and sources gravity (Section 6). *Downward*, gravitational attraction feeds energy back into electromagnetism: the collapse of a positive-mass region converts gravitational potential energy into heat, and that hot dense state drives the electromagnetic processes. Gravity supplies the energy; electromagnetism spends it as structure and light.

Concretely: ordinary gravitational contraction of a positive-mass seed supplies one hot dense epoch, and the released binding energy leaves as light. As the contracting proton–electron matter heats, **deuterons and nuclei** form (∼MeV, RealNucleus), the Coulomb binding energy radiating away (the origin of light) and appearing as the mass defect. On cooling, nuclei capture expanded electrons into **atoms** (∼eV, RealQM), again radiating; at that recombination the matter turns neutral, electromagnetism screens out, and radiation streams free. Only *then* does the weak, slow gravitational aggregation build the cosmic web of Section 6. So the universe's timeline stages the two tiers in order: *charge builds the small while matter is charged; gravity builds the large once it is neutral.*

A bound state survives once the bath can no longer ionise it, so the two electron phases switch on at two temperatures set by their two binding energies, forcing the order plasma → nuclei → atoms (Table 1). The temperature ratio between the epochs ($\sim 10^6$) is just the energy-scale ratio.

epoch	process	temperature
plasma	free ± charges	$> 10^{10}$ K
nuclei	compact e captured → nuclei	$\sim 10^{10}$ K ($kT \sim$MeV)
atoms	expanded e captured → atoms; radiation	$\sim$3000 K ($kT \sim$eV)

Table 1: Two condensation epochs, separated by $\sim 10^6$ in temperature.

The helium fraction: a consistency, not a prediction. The proton–electron picture reproduces the primordial helium; it is worth being exact about what is derived and what is borrowed. The electron has two size-phases — *compact* (inside a nucleus, i.e. inside a neutron, since neutron $= p +$ compact e) and *expanded* (atomic) — interconverted by the two weak processes, compactify ($p + e \to n$) and expand ($n \to p + e$, β-decay). Three quantities enter, and all three are *inputs*:

- E_* — the **energy gap** between the two phases: the energy to compactify an electron, released when it expands. It plays the role the neutron–proton mass difference plays in standard cosmology and is *identified with it*, $E_* \approx 1.29$ MeV. It is an *input*, not a RealNucleus output: its absolute size is the open nuclear-scale problem (Section 11) — RealNucleus fixes dimensionless *ratios*, not the MeV scale.

- T_* — the **transition temperature**, $T_* \sim E_*/k \sim 10^{10}$ K: above it the two phases interconvert freely, below it compactification becomes rare.

- T_f — the **freeze-out temperature** (≈ 0.8 MeV): where the weak (size-changing) processes fall slower than the cosmic expansion and the neutron/proton ratio *locks*. As in standard big-bang nucleosynthesis it is set by the weak-rate-vs-expansion balance, not predicted — a second input.

In equilibrium the neutron/proton ratio is the Boltzmann factor $e^{-E_*/kT}$; at freeze-out it locks at $(n/p)_f = e^{-E_*/kT_f}$; a fraction β-decay back (survival factor d), and essentially all the rest end in He-4:

$$\left(\frac{n}{p}\right)_{\text{nuc}} = \frac{(n/p)_f\, d}{1 + (n/p)_f(1-d)}, \qquad Y = \frac{2\,(n/p)_{\text{nuc}}}{1 + (n/p)_{\text{nuc}}}. \tag{6}$$

With $E_* \approx 1.29$ MeV, $T_f \approx 0.8$ MeV and $d \approx 0.74$: $(n/p)_f = 0.199 \to (n/p)_{\text{nuc}} = 0.140 \to Y = 0.245$, the observed primordial helium. This is a **consistency**, not a new prediction: with the two inputs borrowed — E_* the observed n–p gap, T_f the standard freeze-out — the size-transition picture *reproduces* the standard result. Nothing here is derived; and, notably, RealNucleus's one genuine computed output, the He-4/deuteron binding ratio (13.1 vs 12.7), does *not* enter Y. The content is that the proton–electron reinterpretation is *consistent* with the standard helium calculation, with E_* the same open datum as the nuclear scale.

9 Complexity from the size asymmetry

The world is built not on a charge imbalance (charge is exactly balanced, $\pm e$) but on a *size* imbalance: the proton has a fixed small size (a rigid anchor) while the electron's size is variable, adapting to its well — $\sim$Å in an atom, $\sim$fm in a nuclear cage. The fixed proton supplies the anchors; the variable electron *spans scales* — spreading to bind atoms and molecules, compacting to bind nuclei — producing the nested multi-scale hierarchy that *is* complexity. With a single fixed size only one scale exists and nothing complex is built. The nuclear/atomic separation is thus a **precondition for complexity**, so any universe that supports structure, and hence observers, must exhibit it: an anthropic reason *why* the scales are separated, though not a derivation of *how far*. Notably, since a neutron is structurally $p + e$, it is heavier than a proton *by construction*, so proton and hydrogen stability is automatic — the usual fine-tuning bound "α must not be too large or the proton decays" simply does not arise.

Why the proton must be compact: an energetic argument

The size asymmetry is more than the precondition for complexity: it selects which of the electron's two possible states is bound, and hence whether one obtains an atom, a nucleus, or nothing stable. The decision is purely *energetic* — which configuration is the ground state — not a question of interface stability.

The electron has **two branches**. *Expanded*: a diffuse cloud of size $\sim a_0$ that tapers to zero far away and binds by cusping on a positive charge — the atom, $\sim$eV. *Compact*: a small cloud of size $\sim R_p$ that does *not* taper but hands off to surrounding proton density at nearly zero localisation cost, bound by Coulomb at the small scale — the nuclear glue, $\sim$MeV. Which branch is the ground state is fixed by the proton configuration.

A point proton has no interior. It can hold an electron only *around* it — the electron cusps at the point (maximal density) and spreads outside as the expanded, atomic cloud. It *cannot* hold an electron *inside*, because there is no inside. So a small, point-like proton yields the atom, cleanly and deeply: as $R_p \to 0$ the electron binds at the full -13.6 eV, independent of any boundary condition (forcing the density to zero at the point costs nothing — a measure-zero constraint — and, left free, the electron restores the cusp).

Holding an electron inside needs a cage. The compact branch requires the electron to be *surrounded* by positive charge, so that it can go flat and hand off on all sides at no localisation cost; a lone proton cannot surround it. Two or more protons, close together, can: they form a well with an interior deep enough to make the compact branch the ground state, bound at $\sim$MeV. This is why nuclear binding is collective — and why a single proton, whatever its size, cannot make a nucleus.

The two branches cross at the neutron. A single $p + e$ with the electron squeezed to the proton scale ($R_p \sim a_e$, one proton) is exactly the watershed: too large to be a clean atomic core, too alone to be a deep cage. There the compact branch is *not* the ground state — the expanded branch (a free proton plus electron, i.e. a hydrogen atom) lies lower — so the electron expands and the object decays. This is the free neutron: unstable (β-decay, $\sim$15 min), the decay being nothing but the electron sliding from the compact branch to the expanded one. Its near mass-degeneracy $m_n \approx m_p + m_e$ and the tiny 0.78 MeV release are the signature of sitting on the ridge.

Stable nuclei begin one proton later. Add a second proton and the compact branch drops below the expanded one: $2p + 1e$ (the deuteron) is the first stable electron-glued nucleus, and $4p + 2e$ (the α) is deeply bound — the He-4/deuteron binding ratio coming out 13.1 against an experimental 12.7 from Coulomb geometry alone, no strong force [2]. Free neutron unstable, bound neutron (in a cage of protons) stable: one energetic mechanism, matching nature.

Computations are consistent with this trend. A minimal spherically symmetric model gives the electron bound at -13.6 eV around a point/compact proton, independent of the boundary condition imposed at the kernel. The full free-boundary solver (`molecule_nucleus.js`), run on one and two protons at Neumann continuity, shows the expected *ordering*: a single proton binds the electron in the *expanded* (atomic-size) branch, and adding a second proton and making the protons compact drives the electron to progressively smaller size and deeper binding — the slide from the atomic toward the nuclear branch. The true nuclear (fm) scale lies below the accessible grid resolution ($\sim 0.08\, a_0$), so the absolute nuclear binding and the compact-branch instability of the free neutron are not resolved by these runs; those rest on RealNucleus [2] and on the energetic argument above.

Two levels should be kept apart. That a compact, caged electron is a *well-defined, stable* object at all is a matter of free-boundary stability: the localisation (surface-tension) cost together with non-overlap holds its interface at an energy minimum, so it does not ripple away. But this stabilisation is *universal* — the expanded atomic electron is equally stable — so it cannot by itself decide which object forms. That decision is the *energetic* one: among the boundary-stable configurations, the lowest in energy wins — the expanded atom for a single proton, the compact bound nucleus for a cage of several. Free-boundary stability makes the configurations *possible*; energetics *selects* among them.

10 Relation to ΛCDM and observations

It is worth stating plainly what this model shares with the standard picture, where it differs, and what it does not yet address.

Shared. Ordinary matter is Coulombic and gravitation is attractive between masses; nuclei and atoms form and later assemble into a filamentary large-scale structure of clusters and voids; a helium mass fraction near $Y \approx 0.25$ is accommodated.

Different. There is no initial singularity and no hot Big Bang, hence no cosmic microwave background as a red-shifted afterglow (Section 7); no inflationary expansion of space (the amplification here is of the density *field*, not of space); no dark-matter particle (structure is sourced by the ordinary energy/mass density of Coulombic matter); and no cosmological constant (a repulsive dark-energy sector arises as the negative-mass sign of the gravitational Poisson equation). The neutron is composite $(p + e)$, not elementary, and nuclear binding is Coulombic geometry rather than a strong force.

Not yet addressed. The model is deliberately restricted to Coulomb+Newton and does not, as it stands, account for the redshift–distance relation and the expansion history, the CMB acoustic-peak spectrum under any alternative interpretation, precise large-scale-structure statistics (the two-point correlation and its baryon acoustic feature), or light-element abundances beyond helium. These are the observational fronts on which any serious successor must eventually be tested; here we claim only internal consistency and economy of ingredients, not empirical parity with ΛCDM. What the model offers in exchange is fewer free primitives — two force-constants, no dark-matter species, no separate inflaton, no cosmological constant — at the cost of predictive machinery still to be built (Section 11).

11 Open problems

1. **The proton–electron size difference.** This splits into three questions of unequal status. *(i) The ratio.* The electron is $\sim\alpha^{-2}$ larger than the proton, read here as *geometric*: a caged, zero-localisation electron sits at the proton size R_p and a free one at the atomic a_0, so $a_0/R_p \approx 1/\alpha^2$ (Section 4; no relativity needed). This part the model addresses. *(ii) The absolute scale.* What fixes R_p (∼fm) itself — the same datum as the mass ratio m_p/m_e and the nuclear energy scale: one open number, not derived here (nor in the Standard Model). *(iii) The sign asymmetry.* Why the $+$ sign is the compact one and the $-$ sign the diffuse one *at all*. Pure Coulomb with the Laplacian is charge-conjugation symmetric — swapping $+ \leftrightarrow -$ leaves the equations unchanged — so the asymmetry *cannot* arise from these symmetric ingredients; it requires a symmetry-breaking element. A natural candidate, tied to the fluctuation-spectrum problem below, is a *non-Gaussian (skewed)* potential seed: the super- and sub-level sets of a skewed field are not mirror images — one sign forms sharp, compact pits, the other broad, diffuse swells — a geometric route to one compact and one

extended sign from the *statistics* of the seed rather than a new force. This would supply the *direction* of the asymmetry; its *magnitude* is question (ii).

2. **The fluctuation spectrum and a law of motion.** Here $\rho = \nabla^2\varphi$ is a definition, not yet a dynamics: φ needs its own law of motion, and the seed spectrum is an input. The k^2-weighting makes the amplification short-wavelength dominated, which this model reads as the origin of small-scale matter rather than as a primordial-spectrum prediction (Section 7); comparison with the CMB-inferred $n_s \approx 0.96$ presupposes the Big Bang afterglow interpretation, which the model does not adopt. The status of the CMB in a non-Big-Bang setting is itself an open question, but not one this small-scale-structure model needs to settle.

3. **The mass ratio $m_p/m_e = 1836$, and why it is *not* the nuclear scale.** By the framework's own logic — mass *is* size (small = heavy, E/c^2) — one would expect the mass ratio to set the electron/nucleus *size* ratio, giving a single physical number. It does not. The mass ratio implies a nuclear size $a_0/(m_p/m_e) \approx 29$ fm, whereas the actual scale is ∼2–3 fm ($r_e = \alpha^2 a_0 = 2.8$ fm); equivalently the size ratio $a_0/R_p \approx 2.5 \times 10^4$ ($\approx 1/\alpha^2$) exceeds $m_p/m_e = 1836$ by a factor ∼14. The nucleus is that much *more compact* than the naive size–mass law allows — because the caged electron is flat (zero localisation), held collectively by several protons. **Deriving that factor** — showing the collective flat-caging compacts the electron by exactly this amount — would make m_p/m_e fix the nuclear scale, hence α, and collapse the two numbers into one. That derivation is missing; the gap $1836 \neq 1/\alpha^2$ is the single unsolved step between two numbers and one.

4. **The neutrino.** β-decay's continuous spectrum — the 1930 objection that sank the proton–electron neutron [5]. The "electron expands" step needs a home for that energy and spin; an attempt to have a field carry it came out too weak, so the neutrino is neither derived nor eliminated here.

5. **Precision and scope.** BBN abundances beyond helium, the fluctuation-to-structure statistics, and the extreme regimes (strong-field gravity, the highest energies) lie beyond the Coulomb+Newton scope kept here.

The count of constants. Stripped to essentials, the two forces carry *two coupling constants*, one each, over a unit scale that is mere convention (only ratios are physical): α for Coulomb and G for Newton (dimensionlessly $\alpha_G = Gm_p^2/\hbar c$). Neither is derived here (nor is α in the Standard Model). Everything below them — all of atomic, molecular and chemical physics, and the nuclear binding *ratios* — follows parameter-free. The economy is worth stating against the standard tally: the Standard Model of particle physics carries some **nineteen** free parameters (the gauge couplings, the Higgs sector, and the quark and lepton masses and mixings), and standard cosmology adds several more (the densities of baryons, dark matter and dark energy, the fluctuation amplitude and tilt, and the inflaton potential). Against that, two force-constants over the domain covered here is a striking reduction — with the caveat, below, that closing the remaining gaps is unfinished.

Two honest gaps remain, and it is worth being clear which is which. *Within the Coulomb sector*, the proton/electron energy ratio $m_p/m_e = 1836$ is a further number, distinct from the size ratio $1/\alpha^2 \approx 1.9 \times 10^4$ — off by ∼14. The mass = size reading *wants* the two equal (folding m_p/m_e into α), but the nucleus is ∼14 times more compact than that allows; computing the collective-caging factor ≈10 from the cage geometry would close the gap and is the prize (asserting it incidental would be mere numerology). *Between the sectors*, whether the two force-constants α and G themselves reduce to one — whether Coulomb and Newton are at bottom a single force, so that G is not independent — is the deeper open question this deliberately parallel construction does not settle. Two forces, two constants, and two honest gaps to one: that is the frontier.

12 Summary

In essence: the Laplacian carries two signs and the gradient two signs, and the four combinations realise **two-sign charge at the small scale** (Coulomb → nuclei and atoms) and **two-sign mass/energy at the large scale** (Newton → the cosmic web and its voids).

More fully: on a plain 3D Euclidean space, two forces share one field equation — Poisson's — differing only in a sign and a strength. **Coulomb, on charge**, builds small heavy protons and large light electrons, hence nuclei and atoms by pure Coulomb geometry (RealQM/RealNucleus), with no strong force. **Newton, on the energy of that matter** (mass being energy, E/c^2), gathers it at the large scale, its $\pm$ curvature separating into the cosmic web (positive mass) and dark energy (negative mass); the speed of light limits only the electromagnetic sector. Two mirror forces, two constants (α, G), running in parallel — and two honest questions standing: what sets each constant, and whether the two are at bottom one.

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